

Question 1

(a) There're 2 turns, (8-12, 12-4) and (12-4, 4-8)

L_1, L_2, L_3 means the impatient customer after each shift finishes after one shift finishes.

x_1 , 8am start, finishes at 4pm

x_2 , 12pm start, finishes at 8pm

$$L_1 = 200 - 50x_1 \geq 0$$

$$L_2 = L_1 + 400 - 50x_1 - 50x_2 = 600 - 100x_1 - 50x_2 = 250 - 50x_1 \geq 0 \quad x_1 \leq 5$$

$$L_3 = L_2 + 100 - 50x_2, L_3 = 0 \quad 700 = 100(x_1 + x_2) \Rightarrow x_1 + x_2 = 7$$

$$\text{Cost for pay on tellers} = 400x_1 + 400(x_1 + x_2) + 400x_2 = 800x_1 + 800x_2 = 5600$$

$$\begin{aligned} \text{cost for impatience} &= 10(200 - 50x_1) + 10(600 - 100x_1 - 50x_2) \\ &= 8000 - 1500x_1 - 500x_2 \end{aligned}$$

$$\text{Total cost} = 12600 - 3500 - 1000x_1 = 10100 - 1000x_1$$

$$\begin{aligned} \text{s.t. } x_1 &\leq 4 \\ x_1 + x_2 &= 7 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \min \quad & 10100 - 1000x_1 \\ \text{s.t. } \quad & x_1 + s_1 = 4 \\ & x_1 + x_2 = 7 \\ & x_1, x_2 \geq 0, s_1 \geq 0 \end{aligned}$$

(b) Phase-I: introduce artificial var. a_1

Find an initial BFS

$$\min w = a_1$$

$$\text{s.t. } x_1 + s_1 = 4$$

$$x_1 + x_2 + a_1 = 7$$

$$x_1, x_2, s_1, a_1 \geq 0$$

$$a_1 = 7 - x_1 - x_2$$

$$\text{objective function: } w = 7 - x_1 - x_2$$

Choose s_1, a_1 as basic

x_1, x_2 nonbasic

Initial:

	x_1	x_2	s_1	a_1	RHS
w	1	1	0	0	7
$(x_1) s_1$	1	0	1	0	4
a_1	1	1	0	1	7

Step 1: choose x_1 as entering variable, s_1 as leaving

$$x_1 = 4 - s_1$$

$$\text{obj. } w = 7 - (4 - s_1) - x_2 = 3 + s_1 - x_2$$

x_1 - now not change

$$a_1 \text{ - now } 4 - s_1 + x_2 + a_1 = 7 \quad x_2 - s_1 + a_1 = 3$$

	x_1	x_2	s_1	a_1	RHS
w	0	1	-1	0	3
x_1	1	0	1	0	4
a_1	0	1	-1	1	3

x_1, a_1 basic

x_2, s_1 nonbasic

Step 2: x_2 as entering, a_1 as leaving

$$x_2 = 3 - a_1 + s_1$$

$$\text{obj. } = 3 + s_1 - 3 + a_1 - s_1 = a_1 = w$$

achieve optimal. = 0

$\therefore$ Simplex method phase II applicable.

	x_1	x_2	s_1	a_1	RHS
w	0	0	0	1	0
x_1	1	0	1	0	4
x_2	0	1	-1	1	3

Q1 (b) cont.

Phase I Initial BFS from I: $x_1 = 4$, $x_2 = 3$, $s_1 = 0$

Choose basis: x_1, x_2 $x_1 = 4 - s_1$

$$z = 10/100 - 4000 + 1000s_1$$

$$= 6100 + 1000s_1$$

	x_1	x_2	s_1	RHS	already optimal
z	0	0	-1000	6100	
x_1	1	0	1	4	
x_2	0	1	-1	3	

Optimal value = 6100, where $x_1 = 4$, $x_2 = 3$, $s_1 = 0$

Question 2

(a) assume they consume x_1 steak, x_2 potatoes

objective: minimize $4x_1 + 2x_2$

constraints: $5x_1 + 15x_2 \geq 55$, $20x_1 + 5x_2 \geq 40$, $15x_1 + 3x_2 \geq 60$

In canonical form:

min $4x_1 + 2x_2$;

s.t. $5x_1 + 15x_2 \geq 55$

$20x_1 + 5x_2 \geq 40$

$15x_1 + 3x_2 \geq 60$.

$x_1, x_2 \geq 0$

$$(b) A = \begin{bmatrix} 5 & 15 \\ 20 & 5 \\ 15 & 3 \end{bmatrix} \quad A^T = \begin{bmatrix} 5 & 20 & 15 \\ 15 & 5 & 3 \end{bmatrix}$$

Introduce y_1, y_2, y_3 variables

Dual problem: max $55y_1 + 40y_2 + 60y_3$

s.t. $5y_1 + 20y_2 + 15y_3 \leq 4$,

$15y_1 + 5y_2 + 3y_3 \leq 2$,

$y_1, y_2, y_3 \geq 0$

(c) In (a)

$$x_1^* = 3.5, x_2^* = 2.5$$

Substitute into primal problem

$$5x_1^* + 15x_2^* = 55$$

$$20x_1^* + 5x_2^* \geq 40$$

$$15x_1^* + 3x_2^* = 60$$

Then $y_1^*, y_2^* \geq 0, y_3^* = 0$

And $x_1^*, x_2^* \geq 0$, then:

$$5y_1^* + 20y_2^* + 15y_3^* = 4$$

$$15y_1^* + 5y_2^* + 3y_3^* \leq 2$$

$$\text{Solve: } y_1^* = \frac{2}{25}, y_2^* = \frac{5}{21}$$

Question 3

(a) Introduce variable: y_1, y_2

$$A = \begin{bmatrix} 1 & 3 & -2 & -1 & 4 \\ 2 & 2 & -1 & -5 & 2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ 3 & 2 \\ -2 & -1 \\ -1 & -5 \\ 4 & 2 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 2 & 5 & 0 & 3 & 6 \end{bmatrix}$$

Dual problems: max $3y_1 + 4y_2$

s.t. $-y_1 + 2y_2 \leq 2$

$3y_1 + 2y_2 \leq 5$

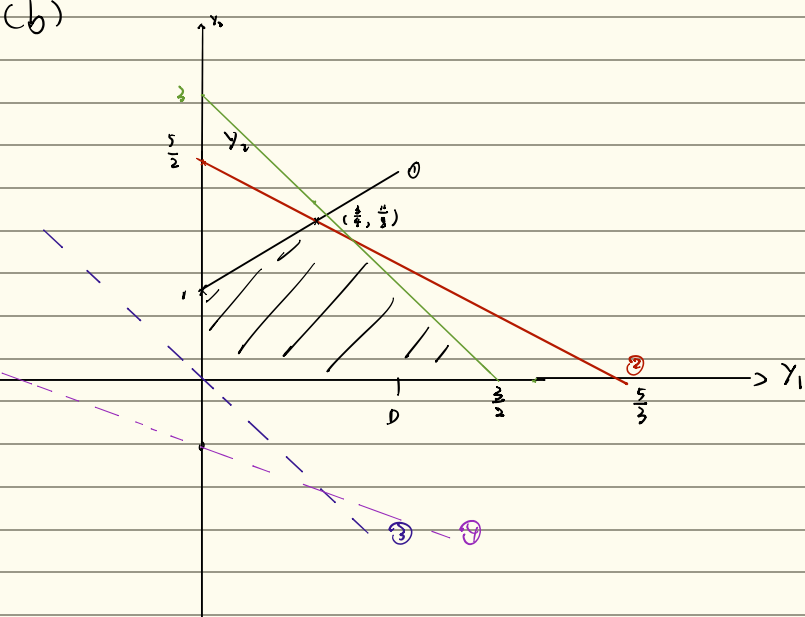
$-2y_1 - y_2 \leq 0$

$-y_1 - 5y_2 \leq 2$

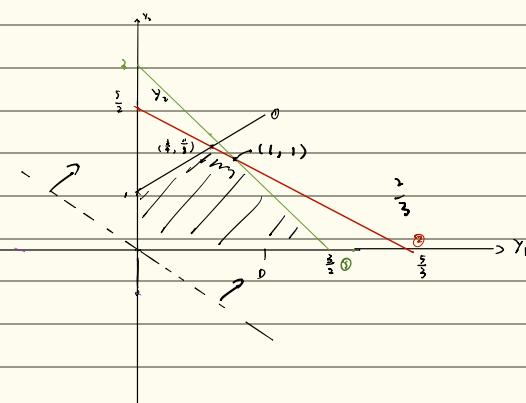
$4y_1 + 2y_2 \leq 6$

$y_1, y_2 \geq 0$

(b)



D: feasible area



Optimal: $(y_1, y_2) = (1, 1)$

c) from (b). $y_1^* = \frac{3}{4}$, $y_2^* = \frac{11}{8}$

Substitute to dual problem.

Optimal value: $3y_1^* + 4y_2^* = \frac{11}{2} + \frac{9}{4} = \frac{21}{4}$
 $-y_1^* + 2y_2^* = 2$
 $3y_1^* + 2y_2^* = 5$
 $-2y_1^* - y_2^* \leq 0$
 $-y_1^* - 5y_2^* \leq 3$
 $4y_1^* + 2y_2^* \leq 6$

And: $y_1^*, y_2^* \geq 0$, so

$-x_1^* + 3x_2^* - 2x_3^* - x_4^* + 4x_5^* = 3$
 $2x_1^* + 2x_2^* - x_3^* - 5x_4^* + 2x_5^* = 4$

$\therefore x_1^* = \frac{3}{4}$, $x_2^* = \frac{5}{4}$

Objective: $\frac{3}{4} \times 2 + \frac{5}{4} \times 5 = \frac{21}{4}$.

$\therefore$ optimal: $(x_1^*, x_2^*, x_3^*, x_4^*, x_5^*)$
 $= (\frac{3}{4}, \frac{5}{4}, 0, 0, 0)$

Question 4

formulate $f(x_2)$.

assume y_1 binary, where $y_1 \in \{0, 1\}$

$y_1 = 0$, $x_2 = 0$, $f(x_2) = 3$

$y_1 = 1$, $x_2 > 0$, $f(x_2) = 10 + 3x_2$

$\therefore f(x_2) = (10 + 3x_2)y_1 + 3(1 - y_1)$
 $= 7y_1 + 3x_2y_1 + 3$

a. Formulate the constraint.

assume y_2 binary

① $2x_1 - x_2 + 3x_3 \geq 6$

If not hold, RHS is -10

② $3x_2 + 4x_3 \leq 9$

if not hold, RHS is 0

assume $y_2 = 1$, ① holds

$y_2 = 0$, ② holds

Assume there's a big-M

constraint:

$$\begin{cases} 2x_1 - x_2 + 3x_3 \geq 6 - (1 - y_2)M \\ 3x_2 + 4x_3 \leq 9 + My_2 \end{cases}$$

b. assume y_3, y_4, y_5, y_6 binary

assume $y_2 = 1$, certain constraints hold. Assume there's big-M

So: $\begin{cases} 5x_1 + 6x_2 - x_4 \geq 9 - (1 - y_2)M \\ 3x_1 + x_2 + 4x_3 \leq 10 + (1 - y_4)M \\ x_2 - x_3 \geq 1 - (1 - y_5)M \\ 3x_3 \geq 5 - (1 - y_6)M \end{cases}$

c. assume y_8 binary, constraint: $(x_1 = 3)$ or $(x_2 \geq 2)$

$$\begin{cases} x_1 \geq 3 - (1 - y_8)M \\ x_1 \leq 3 + (1 - y_8)M \\ x_2 \geq 2 - My_8 \end{cases}$$

d. $x_1, x_2, x_3 \in \mathbb{Z}$

e. $x_4 = 0p_1 + 2p_2 + 4p_3$

$\begin{cases} p_1 + p_2 + p_3 = 1 \\ p_1, p_2, p_3 \text{ binary} \end{cases}$

f. $x_1, x_2, x_3, x_4 \geq 0$

Summarize: $\min x_1 + 3x_2 + 7y_1 + 3y_1x_3 + 3 + 8p_2 + 64p_3$

s.t. $2x_1 - x_2 + 3x_3 \geq 6 - (1 - y_2)M$

$3x_2 + 4x_3 \leq 9 + My_2$

$5x_1 + 6x_2 - 2p_2 - 4p_3 \geq 9 - (1 - y_3)M$

$3x_1 + x_2 + 4x_3 \leq 10 + (1 - y_4)M$

$x_2 - y_5 \geq 1 - (1 - y_5)M$

$3x_3 \geq 5 - (1 - y_6)M$

$x_1 \geq 3 - (1 - y_8)M$

$x_1 \leq 3 + (1 - y_8)M$

$x_2 \geq 2 - My_8$

$y_1, y_2, y_3, y_4, y_5, y_6, y_8, p_1, p_2, p_3$ binary.

$x_1, x_2, x_3 \geq 0$

$x_1, x_2, x_3 \in \mathbb{Z}$

Question 5

(a) assume x_i binary, $i = 1, 2, \dots, 10$

$x_i = 0$, means choose Player i .

objective:

$$\max 12x_1 + 6x_2 + 18x_3 + 26x_4 + 15x_5 + 17x_6 + 9x_7 + 22x_8 + 10x_9 + 12x_{10}$$

constraints: choose 5 players from 10, $\sum_{i=1}^{10} x_i = 5$

at least 1 center: $x_7 + x_8 + x_9 + x_{10} \geq 1$

at least 2 guards: $x_1 + x_2 + x_3 \geq 2$

not more 2 forwards: $x_4 + x_5 + x_6 \leq 2$

Average be at least 6 (total ≥ 30) $4x_1 + 7x_2 + 2x_3 + 10x_4 + 13x_5 + 6x_6 + 8x_7 + 5x_8 + 4x_9 + 7x_{10} \geq 30$

1 wouldn't join if 8 join $x_1 + x_8 \leq 1$ (8, or 1, or neither)

2, 6 will both join: $x_2 - x_6 = 0$ ($x_2 = x_6$)

(b) Either 4, 9 both or 5:

$$x_4 + x_9 = 2 \quad \text{or} \quad x_5 = 1$$

($x_4 + x_9 \geq 2$ and $x_4 + x_9 \leq 2$) or ($x_5 \geq 1$ and $x_5 \leq 1$)

$$\text{Assume } y_1, y_2 \text{ binary} \left\{ \begin{array}{l} x_4 + x_9 \geq 2 - (1-y_1)m, \quad x_4 + x_9 \leq 2 + (1-y_1)m \quad (\text{when } y_1 = 1, \text{ holds}) \\ x_5 \geq 1 - (1-y_2)m, \quad x_5 \leq 1 + (1-y_2)m \quad (\text{when } y_2 = 1, \text{ holds}) \\ y_1 + y_2 \geq 1 \end{array} \right.$$

Average height less than 175

Average Defensive Ability at least 5

$$\left\{ \begin{array}{l} 173x_1 + 180x_2 + 174x_3 + 169x_4 + 181x_5 + 175x_6 + 172x_7 + 178x_8 + 185x_9 + 182x_{10} \leq 175 \times 5 = 875 \\ 6x_1 + 8x_2 + 5x_3 + 3x_4 + 4x_5 + 5x_6 + 9x_7 + 4x_8 + 8x_9 + 6x_{10} \geq 5 \times 5 = 25 \end{array} \right.$$